Paper numbers, training-budget check (exploratory)

This research was supported by data provided by Meituan.

Pre-registration addendum `preregistration-addendum-budget.md`, committed at c217ca4 before any retraining; exploratory, outside every family. Policy 1 of each learned model retrained with three times the budget (600 peaks; Jusup et al. patience 300, at most 3,000 epochs), evaluated at the 30 seeds of policy 1 against the current policy 1 on the same seeds.

Classes, retrained against current policy 1

Seeds 1001, 1006, 1011, ..., 1146 (n = 30), the 16 primary cells, fitted behavior, arm C at 100%. Smallest effects from the pilot; Holm within the E1-shaped and E2-shaped sets of each policy, as a reading device. Contrasts: `b2_contrasts.csv`; classes: `b2_classes.csv`; the DQN's differences: `b2_dqn.csv`.

W1 is W1 over the evening on Q1 cells and W1 over 60 min on Q2 cells; differences are model minus nearest courier, in minutes; lower is better.

input	holm_set	contrasts	established
current	E1-shaped	48	44
current	E2-shaped	80	75
retrained	E1-shaped	48	46
retrained	E2-shaped	80	76

Classes by model (cells):

model	class	retrained	current
auad	does not undo	10	10
auad	does not prevent	6	6
jusup	does not prevent	6	6
jusup	does not undo	5	5
jusup	undoes at no delivery cost	5	5
li	does not undo	10	10
li	prevents at no delivery cost	4	4
li	does not prevent	2	2
xu	does not undo	6	6
xu	does not prevent	4	4
xu	undoes at no delivery cost	2	2
xu	undoes at a delivery cost	2	2
xu	prevents at a delivery cost	1	1
xu	prevents at no delivery cost	1	1

Classes that change: 0 of 64.

None.

Training: the checkpoint kept and the reproduction at peak 200

model	kept	status	budget (peaks; Jusup: most epochs)	reproduces_frozen_policy_1_at_200	epochs_run	best_epoch	frozen_epochs_reproduced
xu	final table (the frozen code saves the table of all peaks)	done	600	True			
li	final actor (the frozen code saves the actor after the last peak)	done	600	True			
dqn	final network (the frozen code saves the network after the last peak)	done	600	True			
auad	theta_best, the theta with the lowest cost on its gate peak (Algorithm 1 lines 13-14)	done	600	True			
jusup	the best validation objective (early stopping keeps the best state)	done	3000		2098	1798	386 of 386

Curves (each model's own objective, and W1 and p90 on the validation seeds, per checkpoint): `b1_curves.csv`.

xu (every 100 peaks)

checkpoint	p90_valid	table_change_frobenius	table_change_max_abs	w1_valid
0	39.240			7.885
100	47.255	16.096	2.002	8.583
200	46.999	9.632	1.894	8.604
300	46.962	7.426	1.802	8.617
400	47.253	5.669	1.055	8.651
500	47.285	5.312	1.509	8.638
600	47.327	4.319	1.167	8.631

li (every 100 peaks)

checkpoint	actor_loss	critic_loss	episode_reward	p90_valid	w1_valid
0				39.728	7.761
100	-0.515	0.536	3344.443	39.726	7.813

checkpoint	actor_loss	critic_loss	episode_reward	p90_valid	w1_valid
200	-0.158	0.383	3306.154	39.731	7.784
300	-0.007	0.382	3638.202	39.786	7.779
400	-0.007	0.284	3518.180	39.734	7.797
500	-0.061	0.474	3927.887	39.748	7.748
600	-0.027	0.442	4107.742	39.727	7.825

dqn (every 100 peaks)

checkpoint	episode_reward	loss	p90_valid	w1_valid
0			71.485	6.876
100	4488.586	0.200	39.627	8.341
200	4784.283	0.187	39.590	8.384
300	4876.488	0.192	39.655	8.390
400	4994.952	0.137	39.609	8.386
500	4821.322	0.201	39.607	8.445
600	4772.414	0.220	39.568	8.392

auad (every 100 peaks)

checkpoint	best_cost	gate_capped_share	gate_cost	p90_valid	train_cost	w1_valid
0				200.264		12.105
100	98912.981	0.560	99134.196	40.205	175934.520	8.447
200	98733.044	0.760	98845.235	40.140	145051.666	8.455
300	98733.044	1.000		40.140	127881.496	8.455
400	98733.044	1.000		40.140	92089.162	8.455
500	98733.044	1.000		40.140	137911.620	8.455
600	97775.503	1.000		40.220	186104.544	8.409

jusup (every 200 epochs)

checkpoint	p90_valid	valid_objective	w1_valid
0	39.213	15.936	8.901
200	38.700	17.917	9.298
400	38.598	18.297	9.536
600	38.556	18.526	9.497
800	38.603	18.674	9.661
1000	38.489	18.864	9.648
1200	38.704	18.955	9.861
1400	38.727	19.041	9.757
1600	38.394	19.095	9.645
1800	38.965	19.510	9.756
2000	38.865	19.561	9.629

Per-policy check of the G8 classes (existing rows)

For each prevent or undo class: each training policy's mean contrast (30 seeds each) against -delta, the pilot's smallest effect of the cell.

model	question	scenario	level	class	delta	policy_1_mean	policy_2_mean	policy_3_mean	policy_4_mean	policy_5_mean	policies_beyond_minus_
li	Q1	burst	loose	prevents at no delivery cost	0.417	-0.801	-0.712	-0.767	-0.739	-0.800	5
li	Q1	burst	normal	prevents at no delivery cost	0.488	-0.661	-0.575	-0.623	-0.612	-0.692	5
xu	Q1	burst	tight	prevents at no delivery cost	0.484	-0.716	-0.744	-0.751	-0.723	-0.685	5
li	Q1	scattered	loose	prevents at no delivery cost	0.394	-0.860	-0.746	-0.785	-0.767	-0.816	5
li	Q1	scattered	normal	prevents at no delivery cost	0.453	-0.750	-0.624	-0.663	-0.649	-0.727	5
xu	Q1	scattered	tight	prevents at a delivery cost	0.482	-0.684	-0.679	-0.741	-0.640	-0.655	5
xu	Q2	meetup	normal	undoes at a delivery cost	0.849	-3.297	-3.749	-3.950	-2.850	-1.119	5
jusup	Q2	mixed	loose	undoes at no delivery cost	1.528	-2.944	-2.757	-2.800	-2.609	-2.941	5

model	question	scenario	level	class	delta	policy_1_mean	policy_2_mean	policy_3_mean	policy_4_mean	policy_5_mean	policies_beyond_minus_
xu	Q2	mixed	loose	undoes at a delivery cost	1.528	-1.786	-1.726	-1.746	-1.676	-1.925	5
jusup	Q2	mixed	normal	undoes at no delivery cost	1.638	-3.002	-2.731	-2.770	-2.683	-2.905	5
xu	Q2	mixed	normal	undoes at no delivery cost	1.638	-2.443	-2.321	-2.318	-2.266	-2.563	5
jusup	Q2	mixed	tight	undoes at no delivery cost	1.612	-1.849	-1.626	-1.502	-1.630	-1.605	3
xu	Q2	mixed	tight	undoes at no delivery cost	1.612	-2.153	-1.971	-2.012	-1.996	-2.290	5
jusup	Q2	three_clusters	loose	undoes at no delivery cost	1.940	-2.853	-2.875	-3.261	-2.955	-2.978	5
jusup	Q2	three_clusters	normal	undoes at no delivery cost	2.049	-2.598	-2.479	-2.908	-2.659	-2.627	5

Auad et al.: the configuration search and the validation cost

alpha	mem	n_step	diverged	theta_best_is_theta0	valid_cost	gate_best_cost	peaks	chosen
1e-07	3000	1000	False	False	88916.600	98733.000	200	True
1e-07	3000	2	False	False	173649.500	92519.600	200	False
1e-06	3000	2	True	True	inf	inf	1	False
1e-06	3000	1000	True	True	inf	inf	2	False
1e-05	3000	2	True	True	inf	inf	1	False
1e-05	3000	1000	True	True	inf	inf	2	False

Validation cost per checkpoint at the selected configuration (mean over the five training seeds and the validation cases): change from 25 to 200 peaks -96.4.

checkpoint	mean_cost	seed_1	seed_2	seed_3	seed_4	seed_5
25	96050.800	89398.570	122906.240	89185.530	90072.280	88691.380
50	96039.310	89341.130	122906.240	89185.530	90072.280	88691.380
75	96039.310	89341.130	122906.240	89185.530	90072.280	88691.380
100	96060.500	89447.050	122906.240	89185.530	90072.280	88691.380
125	96060.500	89447.050	122906.240	89185.530	90072.280	88691.380
150	95954.410	88916.590	122906.240	89185.530	90072.280	88691.380
175	95954.410	88916.590	122906.240	89185.530	90072.280	88691.380
200	95954.410	88916.590	122906.240	89185.530	90072.280	88691.380

This research was supported by data provided by Meituan.